

# The Forward Process

DATA 443/643 · Week 12, note 12.1

Week 12

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Ideas from Stanford CS231N 2025 L14 (corrupt  $x$ ; jump to  $x_t$  in one shot). Original slides; that course is not copied.

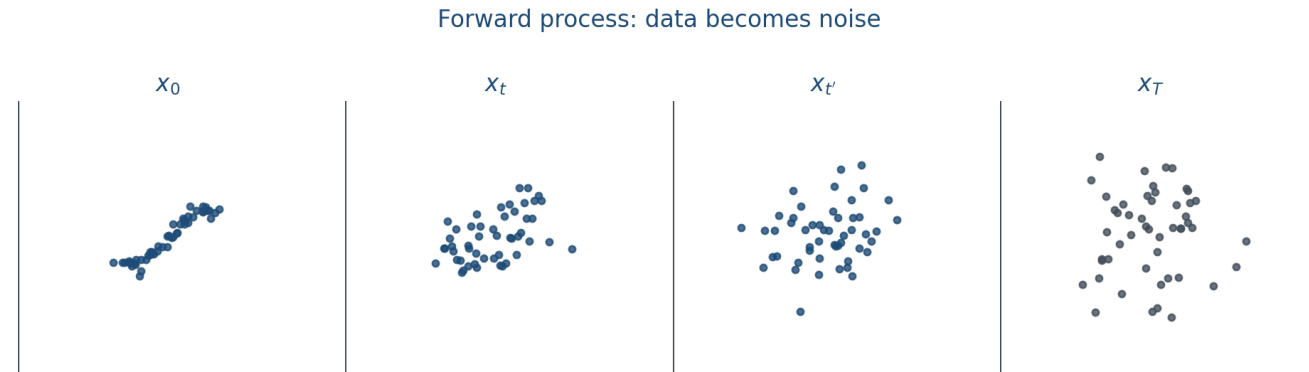
# Lecture plan

- 1 A fixed kernel. No discriminator.
- 2  $\alpha_t$  and  $\alpha\text{-bar}_t$
- 3 The one-line reparameterization
- 4 Print  $\alpha\text{-bar}$  at 1,  $T/2$ ,  $T$

Key goal: training samples  $t$  and jumps. You do not walk  $1 \dots t$ .

# Destroy the sample on purpose

- $\alpha_t = 1 - \beta_t$ ,  $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ .
- Markov:  
 $q(x_t | x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t} x_{t-1}, \beta_t I)$ .
- The forward process is not learned.



# Closed form, one Gaussian

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon$$

$\varepsilon \sim \mathcal{N}(0, I)$ .

$\bar{\alpha}_t = 1$  keeps  $x_0$ . Zero keeps  $\varepsilon$ .

That one-liner is the whole point of this note.

# A scalar path

- $x_0 = 1, \varepsilon = 2$ .
- At  $\bar{\alpha} = 0.64, x_t = 2.0$ : more noise than signal.
- Lab 12: same square roots on a 2-D vector.

$x_0 = 1, \varepsilon = 2$ . At  $\bar{\alpha} = 0.64$  you already sit at 2.0.

| $\bar{\alpha}_t$ | $\sqrt{\bar{\alpha}} x_0$ | $x_t$ |
|------------------|---------------------------|-------|
| 1                | 1.0                       | 1.0   |
| 0.64             | 0.8                       | 2.0   |
| 0                | 0                         | 2.0   |

# Do not simulate the chain at train time

## Markov loop

Walk  $t$  hops of  $q(x_s | x_{s-1})$ . Slow, and unnecessary for the loss.

## Jump

Sample  $t$ , draw  $\epsilon$ , form  $x_t$ . CS231N L14: noise levels, then learn to undo a bit.

# What to take from this lecture

## Three names

$\beta_t$ ,  $\alpha_t$ ,  $\bar{\alpha}_t$ . Write them every time.

## Too fast

Every  $\beta=0.9$ : early  $t$  is already junk.

## Lab 12

Print  $\bar{\alpha}$  at 1,  $T/2$ ,  $T$ . They should fall.